

INDEX

S.No.	Date	Title
1.	29/1/26	Azure Devops Environment Setup.
2.	22/1/26	Azure Devops Project Setup and User Story Management.
3.	29/1/26	Setting Up Epics, Features, And User Stories for Project Planning.
4.	5/2/26	Sprint Planning.
5.	12/2/26	Poker Estimation.
6.	19/2/26	Designing Class and Sequence Diagrams for Project Architecture.
7.	26/2/26	Designing Architectural and ER Diagrams for Project Structure.
8.	5/3/26	Testing – Test Plans and Test Cases.
9.	12/3/26	Load Testing and Pipelines.
10.	4/2/26	GitHub: Project Structure & Naming Conventions.

I N D E X

ARUN PRATHI K

STD.: _____ SEC.: _____ ROLL NO.: _____ SUB.: _____

Date	Title	Page No.	Teacher's Sign / Remarks
8/1	Azure devops environment setup.		P/
22/1	Azure devops project setup and user story management.		P/
29/1	setting up epics, features, and user stories for project planning.		P/
5/2	sprint planning		P/
12/2	Poker estimation		P/
19/2	Designing class and sequence diagrams for project architecture.		P/
26/2	Designing architectural and ER diagrams for project structure.		P/
5/3	Testing - Test plans and Test Cases.		P/
12/3	Load Testing and pipelines.		P/
4/2	Git hub: Project structure & Naming conventions.		P/

AZURE DEVOPS ENVIRONMENT SETUP

EXP NO: 1

AIM:

To setup and access the azure devops environment by creating an organisation through the azure portal.

INSTALLATION:

- * Open your browser and go to azure website.
- * Sign in using your microsoft account credentials.
- * If you don't have a microsoft account, create one using the link;
" <https://sign-up.live.com/?lic=1> "
- * Open devops environment in the azure devops organisation in the search tab.
- * click on the link and create an organisation to navigate to the azure devops organisation home page.

RESULT:

Successfully accessed the azure devops environment and created new organisation through the azure portal.

EXP No: 2

AZURE DEVOPS PROJECT SETUP AND WORK STORY MANAGEMENT

DATE:

AIM:

To setup an azure devops projects for efficient collaboration and agile work management.

INSTRUCTIONS:

- Create an azure account
- Setup the organisation
- On the organisation's home page, click on the new project button.
- Enter the project name, description, and visibility options.
- Open the boards option from the left navigation menu to access the main boards page.
- click add a work item or the + button.
- From the drop down, choose user story.
- A form will open where you can enter all required details for the new user story.

RESULT:

Successfully created an azure devops project with user story management and agile work flow.

DATE:

SETTING UP EPICS FEATURES AND USER

APP NO: 3

STORIES FOR PROJECT PLANNING

AIM:

To learn about how to create epics, user story, features, backlogs from assigned project (To do list app with remainders)

Instructions:

- Define epics like task management, remainders and user login for the to do app
- Break epics into features such as add task, edit task, etc.
- User story - As a user,
I want to add a task,
so that I don't forget my work
- Add acceptance criteria
- Prioritize features.

RESULT:

Thus the epic, features, user stories and tasks have been created successfully.

DATE:

EXP NO: 4

SPRINT PLANNING

AIM:

To assign user story to specific sprint for the to do list app.

SPRINT PLANNING:

- Define sprint goal clearly
- Select user story from product back log.
- Break user stories into smaller task.
- Estimate effort for each task
- Assign task and confirm team capacity.

RESULT:

~~The~~ sprints are created for the to do app.

DATE:

EXP NO: 5

POKER ESTIMATION

AIM:

create poker estimation for user stories of a to do list app.

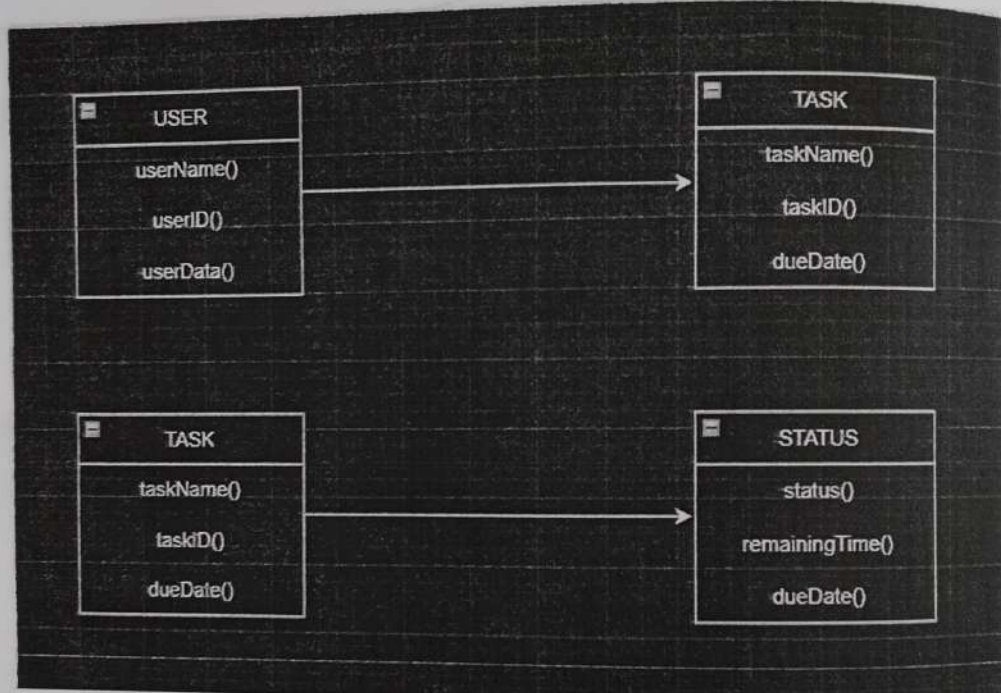
STEPS:

- The team uses cards to estimate efforts.
- Product owner explains the user story.
- All cards revealed simultaneously.
- Discuss differences and repeat until consensus is reached.

RESULT:

~~The~~ estimation is created for the project using ~~poker~~ estimation.

Exp No



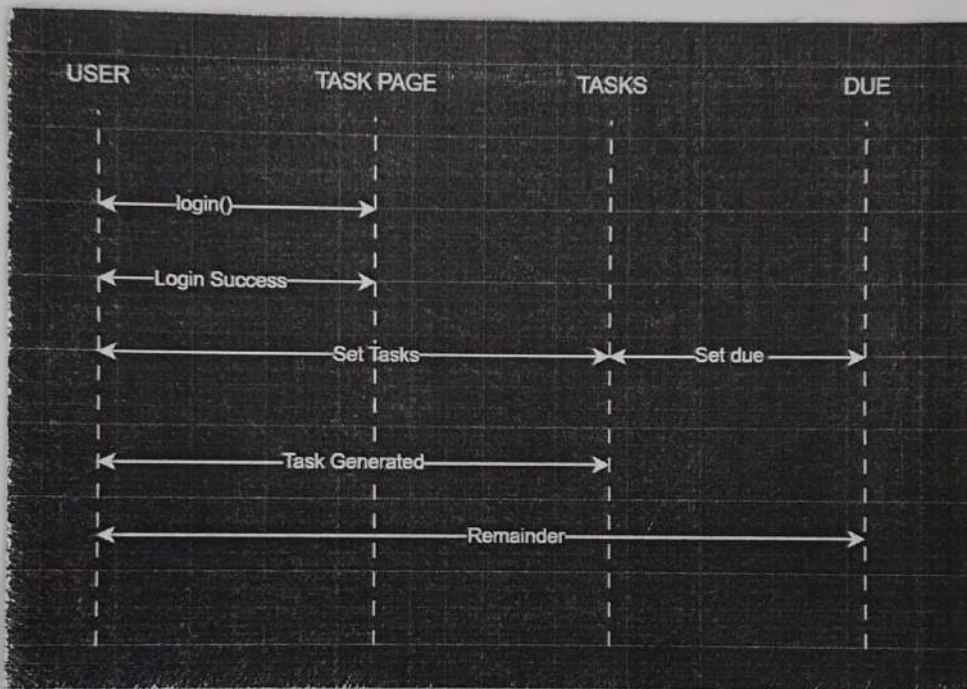
DATE: _____
DESIGNING CLASS AND SEQUENCE DIAGRAMS

Exp No: 6

FOR PROJECT ARCHITECTURE

AIM:

To design a class diagram and sequence diagram for the given project.



RESULT:

The class diagram and sequence diagram is designed successfully for the TO do list app with remainders.

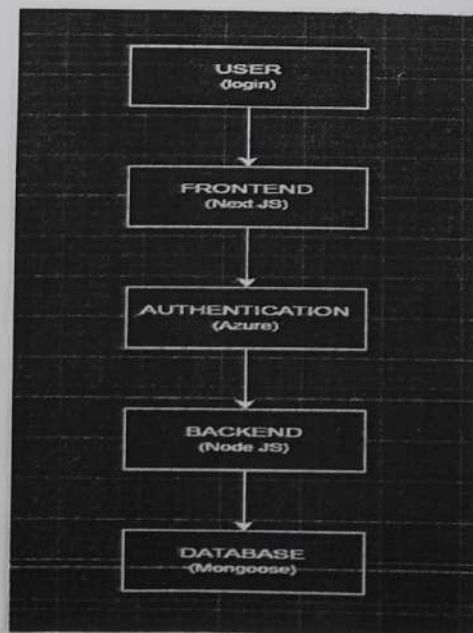
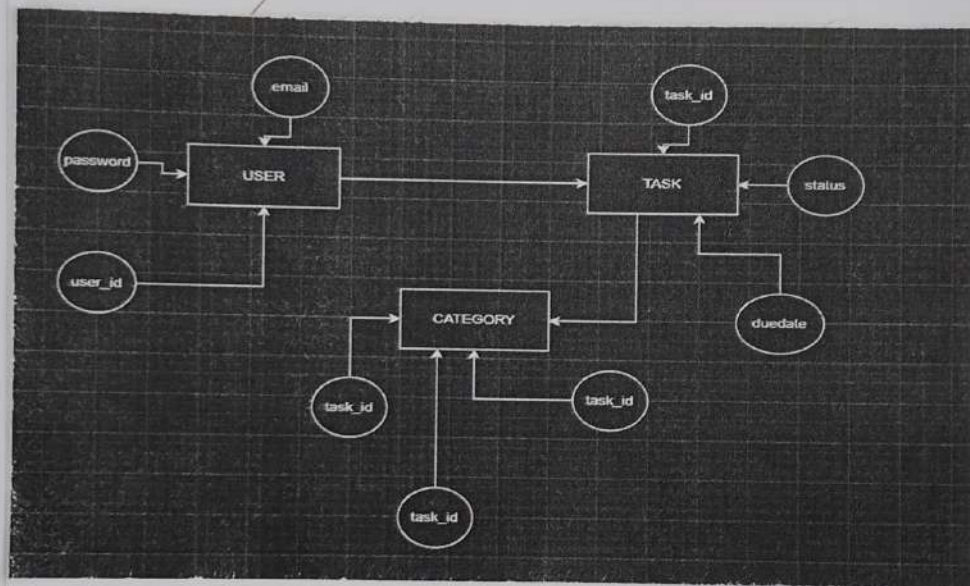
DATE:

DESIGNING ER AND ARCHITECTURAL DIAGRAMS FOR PROJECT STRUCTURE

EXP NO: 7

AIM:

TO design an architectural diagram and ER diagram for the given project.



RESULT:

The architectural diagram and ER diagram are designed successfully for the To do list app with reminders.

TESTING - TEST PLANS AND TEST CASES

AIM:

Test plans and test cases and write two cases for atleast 5 user stories showcasing the happy path and error scenarios in azure devops platform.

Test case design procedure:

- * Identify the features involved in the todo app like login, reminders, etc..
- * Understand the user actions
- * Define the expected behaviour for each test case.
- * Note the constraints like date, time, etc..

Test scenario identification:

- * Task operations
- * Remainder of actions
- * User interface actions
- * Error / edge cases

Test case design:

- * Write steps to perform the actions.
- * Give input values
- * Define the expected output for each input
- * Assign a unique id for each test case

Test case execution:

- * Run each test case
- * observe actual output
- * Compare with expected
- * Mark them as pass or fail based on the actual output.

Defect reporting:

- * Identify ~~the~~ failed cases
- * Describe the issue
- * Add the expected and actual output's

Comparison:

- * Assign severity.

Test coverage check:

- * Cover all features
- * Include edge cases.
- * validate remainder timing.
- * Execute/ ensure UI corrections.

Test plan designing procedure:

objectives:

- * verify app features work correctly
- * Ensure remainder triggers properly.
- * Validate ~~user~~ inputs.

Scope:

- Add, edit and delete task.
- Set and manage reminders
- Notification system.
- User interface

Test strategy:

- check if each feature works correctly
- Adding a task should always add it
- Deleting should remove it
- Reminder should be saved properly.

Negative testing:

- Check how the app behaves with wrong input.
- For ex, when the user does not enter the input and generates the task, or setting reminders in the past or using invalid time format.

Reminder timing check:

- Verify reminder triggers at correct time.
- Ensure time accuracy.
- Check delays or early execution.
- Confirm reliability of notification.

User stories:

- As a user, I want to add a task so that I can keep track of my work.
- As a user, I want to delete a task so that I can remove completed or unwanted tasks.
- As a user, I want to mark a task as complete so that I can know which tasks are finished.
- As a user, I want to set a reminder for a task so that I don't forget it.
- As a user, I want to receive a notification so that I am alerted at correct time.

RESULT:

Hence the test case and test plan are created for the user stories using azure devops.

DATE:

LOAD TESTING AND PIPELINES

EXP NO. 9

AIM:

To create an azure load testing resource and run a load test to evaluate the performance of a target end point and to create and demonstrate an azure devops pipeline for automating application builds, tests and deployment.

9A) Load testing:

The load test was executed in azure load testing and metrics such as virtual users and response time were observed.

9B) Application UI:

The to do app application interface was used as a target system for performing load testing.

pipeline setup:

The azure devops pipeline was configured by connecting github repository and defining build and test steps.

yaml code:

trigger:
- main

Pool:

Yml Image : ubuntu - latest

Steps:

step 1: checkout the code from github
- checkout: self

step 2: set up python environment
- task: use Python version @ 0
inputs:

version spec: "3x"

~~display name~~: "set up Python"

step 3: Install dependencies from correct path
- script: |

python -m install

- upgrade pip

pip install -r project/requirements.txt

display name: "Install dependencies"

step 4: ~~Run~~ a simple platform script to check
the environment.

- script: |

python -c "print ('Good Morning')"

The pipeline was successfully executed in azure devops, showing build status and job execution details.

RESULT:

Successfully created azure load testing resource and executed a load test and performance of specified end point are also demonstrated pipeline in azure devops.

EXP NO: 10

GITHUB: PROJECT STRUCTURE AND NAMING CONVENTIONS

AIM:

To provide a clear organised view of projects folder structure and file naming conventions, helping contributors and users easily understand, navigate and extend the to do app with ~~remainders~~ project.

PROCEDURE:

- * Create a repository on github
- * Open the terminal and navigate to the project folder.
- * Initialize git
- * Add and commit files
- * Connect to github and push the files.

RESULT:

The github repository clearly displays the original project structure and making it easy for users and contributors to understand and navigate the database.